
Introduction and Mathematical Foundations

CS5487: Machine Learning: Principles and Practice

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Semester A, 2026/27

What is machine learning?

CS5487 ■ Semester A, 2026/27

What is machine learning?

Machine learning uses **data** to learn useful patterns, models, or representations that can be applied to new observations.

Data $\longrightarrow$ **Learning** $\longrightarrow$ **Model or representation**

Machine learning can be used to:

- ▶ predict a category, such as *spam or not spam*;
- ▶ predict a numerical value, such as *the price of a house*;
- ▶ discover useful structure in data.

The central goal is generalization: what we learn from observed data should remain useful on unseen data.

A concrete learning problem

Suppose we want an email system to identify **spam** automatically. We already have many past emails labeled as *spam* or *not spam*.

Question

Can we learn from these past examples and correctly classify a new email that we have never seen before?

Input x : a feature vector describing an email, such as information about its words, sender, and links.

Target $y \in \{0, 1\}$: not spam or spam.

Errors Spam may reach the inbox, or a legitimate email may be incorrectly sent to spam.

Check your understanding. How can we tell whether the learned rule works well on new emails?

Common learning settings

Suppose we have n training examples, indexed by $i = 1, \dots, n$.

Supervised learning

Observe input–target pairs (x_i, y_i) .

- ▶ **Classification:** predict a category
e.g., spam or not spam.
- ▶ **Regression:** predict a number
e.g., the price of a house.
- ▶ Goal: predict y for a new x .

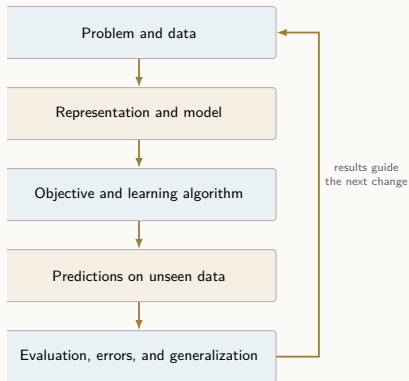
Unsupervised learning

Observe inputs $x_1, \dots, x_n$ without target labels.

- ▶ **Clustering:** discover groups.
- ▶ **Density estimation:** model how data are distributed.
- ▶ **Dimensionality reduction:** find compact representations.

The key distinction is whether target labels are available during learning.

A machine-learning study is a connected system



A reliable ML study connects the problem, data, learning process, predictions, and evaluation.

What should you be able to do by the end?

By the end of this course, you should be able to:

1. **Understand and derive** the principles behind common machine-learning methods;
2. **Implement** selected methods to solve practical problems;
3. **Compare and evaluate** methods using mathematical reasoning and experimental evidence;
4. **Design** an appropriate machine-learning solution for a given problem.

The goal is not just to know algorithm names, but to understand when a method is appropriate, why it works, and how to evaluate it.

Course information

CS5487 ■ Semester A, 2026/27

Course information and learning format

Course	CS5487: Machine Learning: Principles and Practice ■ 3 credit units
Instructor	Kangning Cui ■ AC4-422 ■ kangning.cui@cityu-dg.edu.cn
Teaching Assistant	Qiuyu Chen ■ cqycsu@outlook.com
Monday	AC5-514 ■ Lecture 9:30–11:20 AM ■ Tutorial 11:30 AM–12:20 PM

Lecture Explain assumptions and derive the main equations.

Tutorial Practise calculations, interpretation, and selected implementation.

Assignments Theory, derivation, model analysis, and/or programming.

Group Project Define, compare, evaluate, and explain a complete ML study.

Assessment

Assessment	Weight	Information
Assignments	20%	Theory, derivation, model analysis, and/or programming exercises.
Midterm	20%	Foundations and main methods from the first half.
Group Project	30%	Problem, experiments, report, and presentation.
Final Examination	30%	Understanding and application across the course.

Exact dates and detailed instructions will be posted on Canvas.

Semester roadmap: foundations and probabilistic learning

Week	Topic	Connection
1	Introduction and review	Language for uncertainty and geometry
2	Parameter estimation	Data determine model parameters
3	Bayesian estimation	Prior information combines with data
4	Parametric clustering	Hidden labels and mixture models
5	Non-parametric clustering	Local density and graph structure
6	National Holiday / Revision	No class; midterm revision
7	Midterm & Bayesian classifiers	Decision theory and probabilistic classification

Semester roadmap: modern ML to nonlinear learning

Week	Topic	Connection
8	Modern ML & project directions	Modern models, datasets, and project planning
9	Linear dimensionality reduction	Covariance, projection, and SVD lead to PCA
10	Linear classifiers I	Likelihood, loss, and linear prediction
11	Linear classifiers II	Margin selects a robust separator
12	Kernel methods	Inner products support nonlinear representations
13	Nonlinear dimensionality reduction	Preserve curved or local geometry
14	Project presentations	Explain evidence, errors, and limitations

Today's learning objectives

By the end of this lecture, you should be able to:

1. distinguish the roles of a model, objective, learning algorithm, and evaluation;
2. explain training, validation, testing, and basic classifier evaluation;
3. use joint, marginal, conditional probability and Bayes' rule;
4. interpret expectation, variance, covariance, Gaussian geometry, projections, eigenvectors, and SVD.

Model, objective, and learning algorithm are different

Model A prediction rule $f_{\theta}(x)$.

Parameters θ contains adjustable values that determine how the model behaves.

Objective $L(\theta)$ measures prediction error on the training task; smaller values represent better solutions.

Learning algorithm A procedure that uses training data to choose $\hat{\theta}$.

Prediction $\hat{y} = f_{\hat{\theta}}(x)$ is the model's predicted output for a new input.

$$\hat{\theta} = \arg \min_{\theta} L(\theta) \quad \implies \quad \hat{y} = f_{\hat{\theta}}(x).$$

Do not confuse these. The model makes predictions; the learning algorithm chooses its parameters.

Training, validation, and test data have different jobs

Data	Used for	Not used for
Training	Fit model parameters and learn preprocessing steps.	Final performance claims.
Validation	Choose between candidate models and model settings.	Final performance claims.
Test	Evaluate the completed system on unseen data.	Model or preprocessing choices.

Common mistake

Data leakage occurs when information from validation or test data is used during model fitting or preprocessing. For example, do not compute normalization statistics using the full dataset.

The test set is not part of model development.

Every prediction belongs to one of four groups

Suppose the test set contains 100 spam emails and 900 legitimate emails.

	Predicted spam	Predicted not spam
Actually spam	TP = 90	FN = 10
Actually not spam	FP = 45	TN = 855

- ▶ TP: spam correctly detected.
- ▶ FN: spam missed by the filter.
- ▶ FP: legitimate email incorrectly sent to spam.
- ▶ TN: legitimate email correctly kept.

The denominator tells us which question a metric answers

Treat **spam** as the positive class.

Measure	Formula	Plain-language question
Sensitivity / Recall	$\frac{TP}{TP + FN}$	Among actual positives, how many did we find?
Specificity	$\frac{TN}{TN + FP}$	Among actual negatives, how many did we reject?
Precision	$\frac{TP}{TP + FP}$	Among predicted positives, how many were correct?
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Among all examples, how many were correct?

Do not confuse these. Sensitivity is recall, not precision. Accuracy can hide important errors when one class is rare.

Probability as a language for uncertainty

CS5487 ■ Semester A, 2026/27

Random variables and distributions

A random variable assigns a numerical value to an uncertain outcome.

- ▶ **Discrete:** $Y = 1$ if an email is spam and $Y = 0$ otherwise. Its probability mass function is $p_Y(y) = \mathbb{P}(Y = y)$.
- ▶ **Continuous:** X may represent a numerical feature such as email length. Its density $f_X(x)$ satisfies

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

Total probability equals one:

$$\sum_x p_X(x) = 1 \quad \text{or} \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Do not confuse these. For a continuous variable, $f_X(x)$ is a density, not the probability of observing exactly x .

Expectation: a probability-weighted average

$$\mathbb{E}[X] = \sum_x x p_X(x) \quad \text{discrete,}$$

$$\mathbb{E}[X] = \int x f_X(x) dx \quad \text{continuous.}$$

For any function g ,

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x) \quad \text{or} \quad \int g(x) f_X(x) dx.$$

Linearity of expectation

$\mathbb{E}[aX + b] = \mathbb{E}[aX] + \mathbb{E}[b] = a\mathbb{E}[X] + b$. Independence is **not** required.

Variance measures spread around the mean

Let $\mu = \mathbb{E}[X]$. Then

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

Expanding the square gives a useful computational identity:

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \boxed{\mathbb{E}[X^2] - \mathbb{E}[X]^2}.\end{aligned}$$

Also, $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

Joint and marginal distributions

Let

$S = 1$: the email is actually spam, $A = 1$: the filter predicts spam.

The joint distribution describes S and A together.

	$A = 0$	$A = 1$	$p_S(s)$
$S = 0$	0.855	0.045	0.900
$S = 1$	0.010	0.090	0.100
$p_A(a)$	0.865	0.135	1

A **marginal distribution** is obtained by summing out the other variable:

$$\mathbb{P}(S = 1) = 0.010 + 0.090 = 0.100.$$

Conditional probability

Conditioning means focusing on cases where a given event has occurred:

$$p(x | y) = \frac{p(x, y)}{p(y)}, \quad p(y) > 0.$$

Equivalently,

$$p(x, y) = p(x | y)p(y).$$

Spam example

$$\mathbb{P}(A = 1 | S = 1) = \frac{0.090}{0.100} = 0.90$$

Recall: among actual spam, how many are detected?

$$\mathbb{P}(S = 1 | A = 1) = \frac{0.090}{0.135} = 0.667$$

Precision: among predicted spam, how many are actually spam?

Do not confuse these. $\mathbb{P}(A | S)$ and $\mathbb{P}(S | A)$ answer different questions.

Before Bayes' rule: what do the four terms mean?

Let D mean that a person has a disease, and let $+$ denote a positive test result.

Prior $\mathbb{P}(D)$: how likely is the disease before seeing the test result?

Likelihood $\mathbb{P}(+ | D)$: how likely is a positive result if the disease is present?

Posterior $\mathbb{P}(D | +)$: how likely is the disease after observing a positive result?

Evidence $\mathbb{P}(+)$: how likely is a positive result overall?

Do not confuse these. $\mathbb{P}(+ | D)$ and $\mathbb{P}(D | +)$ answer different questions.

Bayes' rule: reversing a condition

From the product rule,

$$\mathbb{P}(D \mid +)\mathbb{P}(+) = \mathbb{P}(+ \mid D)\mathbb{P}(D).$$

Therefore,

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\mathbb{P}(D)}{\mathbb{P}(+)}.$$

A positive result can occur in two ways:

$$\mathbb{P}(+) = \mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid \neg D)\mathbb{P}(\neg D).$$

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

Bayes' rule converts “probability of the evidence given a hypothesis” into “probability of the hypothesis given the evidence.”

Example: a medical test

Suppose:

- ▶ the disease occurs in 1% of people: $\mathbb{P}(D) = 0.01$;
- ▶ sensitivity is 95%: $\mathbb{P}(+ | D) = 0.95$;
- ▶ the false-positive rate is 5%: $\mathbb{P}(+ | \neg D) = 0.05$.

$$\begin{aligned}\mathbb{P}(D | +) &= \frac{\mathbb{P}(+ | D)\mathbb{P}(D)}{\mathbb{P}(+ | D)\mathbb{P}(D) + \mathbb{P}(+ | \neg D)\mathbb{P}(\neg D)} \\ &= \frac{0.95(0.01)}{0.95(0.01) + 0.05(0.99)} \\ &\approx 0.161.\end{aligned}$$

A positive test does not imply a 95% chance of disease. The prevalence of the disease also matters.

Understanding check

A classifier has 99% sensitivity and 99% specificity, but the event occurs in only 0.1% of cases.

Check your understanding. After a positive prediction, is the event more likely to be truly present or absent?

Common mistake

Do not confuse sensitivity with precision. Rare positive classes can still produce many false positives.

$$\mathbb{P}(\text{event} \mid +) = \frac{0.99(0.001)}{0.99(0.001) + 0.01(0.999)} \approx 0.09.$$

Only about 9% of positive predictions are truly positive.

Common distributions and covariance

CS5487 ■ Semester A, 2026/27

Three useful discrete distributions

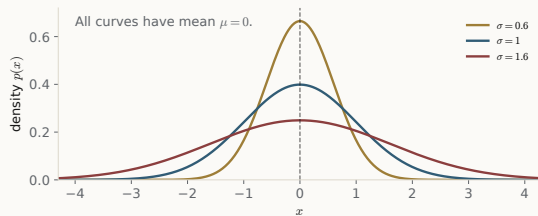
Distribution	Model	Meaning / example
Bernoulli	$p(x) = \pi^x (1 - \pi)^{1-x}$	One binary outcome; $\pi = \mathbb{P}(X = 1)$.
Binomial	$p(k) = \binom{n}{k} \pi^k (1 - \pi)^{n-k}$	Number k of successes in n independent trials.
Poisson	$p(k) = e^{-\lambda} \lambda^k / k!$	Number k of events; λ is the expected count.

$$\mathbb{E}[X]_{\text{Bern}} = \pi, \quad \mathbb{E}[K]_{\text{Bin}} = n\pi, \quad \mathbb{E}[K]_{\text{Pois}} = \lambda.$$

The univariate Gaussian

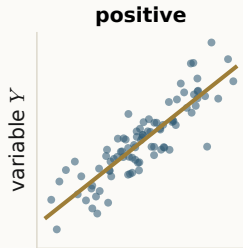
$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right].$$

- ▶ μ is the mean and determines the center.
- ▶ $\sigma > 0$ is the standard deviation; σ^2 is the variance.
- ▶ Density decreases as the squared distance from μ increases.

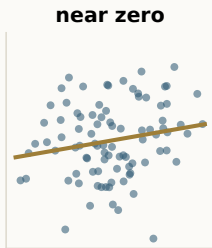


Do two variables move together?

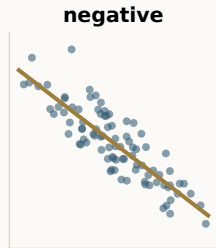
Consider height X and weight Y measured for many people. Each point represents one person.



increase together



no clear linear direction



one rises as the other falls

Covariance is positive when the two variables tend to rise together and negative when one tends to fall as the other rises.

Covariance measures joint variation

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)].$$

- ▶ Positive: large X tends to occur with large Y .
- ▶ Negative: large X tends to occur with small Y .
- ▶ Zero: no *linear* association; not necessarily independence.

The product $(X - \mathbb{E}X)(Y - \mathbb{E}Y)$ is positive when both variables are above their means or both are below their means.

Do not confuse these. Covariance depends on measurement units. Correlation divides covariance by both standard deviations, so it is unit-free and lies between -1 and 1 .

Reading a covariance matrix

For two features X_1 and X_2 ,

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \end{bmatrix}.$$

Their covariance matrix is

$$\boldsymbol{\Sigma} = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \text{Var}(X_2) \end{bmatrix}.$$

- ▶ **Diagonal entries:** how much each feature varies.
- ▶ **Off-diagonal entries:** how the two features vary together.

A covariance matrix stores variances and pairwise covariances in one object.

From one Gaussian to many dimensions

For one variable, a Gaussian is described by

mean μ and variance σ^2 .

For several features, we use

mean vector μ and covariance matrix Σ .

The multivariate Gaussian has the form

$$p(x) \propto \exp \left[-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right].$$

- μ determines the **center**. Σ determines the **spread and orientation**.

The covariance matrix turns a one-dimensional Gaussian curve into a multidimensional shape.

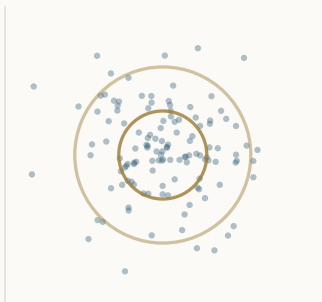
Ellipse width and height show spread

For a fixed constant $c > 0$, points satisfying

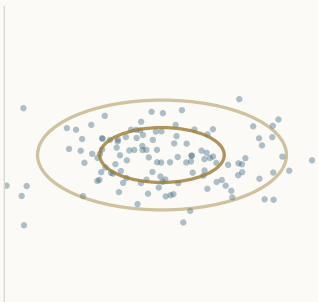
$$(x - \mu)^T \Sigma^{-1} (x - \mu) = c$$

lie on the same Gaussian density contour.

same spread



more horizontal spread

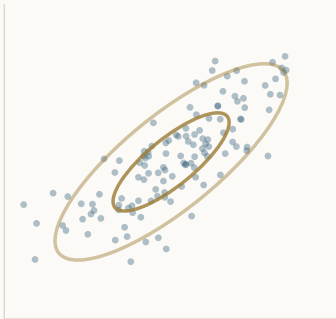


more vertical spread

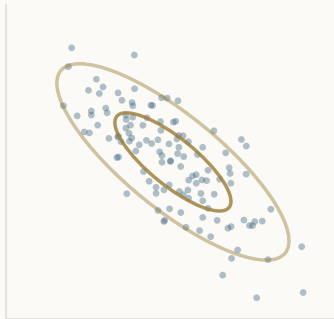


Ellipse tilt shows joint movement

positive covariance



negative covariance



Shape shows spread. Tilt shows the sign of the linear relationship.

What the covariance ellipses tell us so far

1. The center is the mean μ .
2. A long axis marks a direction with greater variation.
3. Tilt records joint linear movement between features.
4. Equal-density points have the same covariance-adjusted distance from the center.

**We still need a mathematical way to find the ellipse's special directions.
That is the role of eigenvectors.**

Linear algebra for learning

CS5487 ■ Semester A, 2026/27

Dimensions tell us what a matrix represents

Let n be the number of samples and d the number of features.

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix} \in \mathbb{R}^{n \times d}.$$

Each row is one sample; each column is one feature.

$$\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{d \times d}, \quad \mathbf{X} \mathbf{X}^\top \in \mathbb{R}^{n \times n}.$$

- ▶ $\mathbf{X}^\top \mathbf{X}$ relates **features to features**.
- ▶ $\mathbf{X} \mathbf{X}^\top$ relates **samples to samples**.

Check your understanding. Which matrix would naturally appear when studying relationships between features?

Dot product and projection

Let $\mathbf{u}$ be a unit direction:

$$\|\mathbf{u}\|_2 = 1.$$

The dot product

$$z = \mathbf{u}^\top \mathbf{x}$$

gives the coordinate of $\mathbf{x}$ along direction $\mathbf{u}$.

The corresponding projected vector is

$$\hat{\mathbf{x}} = (\mathbf{u}^\top \mathbf{x})\mathbf{u}.$$

Projection describes a point by how far it lies along a chosen direction.

What does an eigenvector mean?

A matrix A transforms a vector:

$$u \longrightarrow Au.$$

Usually, the direction changes.

An **eigenvector** is a special direction satisfying

$$Au = \lambda u.$$

- ▶ u : a direction preserved by the transformation.
- ▶ λ : how strongly that direction is scaled.

Eigenvectors reveal the natural directions of a matrix.

From covariance eigenvectors to PCA

For a covariance matrix,

$$\Sigma \mathbf{u} = \lambda \mathbf{u}.$$

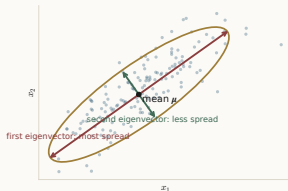
This has a useful interpretation:

- ▶ $\mathbf{u}$ gives a direction of data variation;
- ▶ λ gives the variance along that direction.

A larger eigenvalue means greater spread.

Principal component analysis (PCA)

keeps the directions with the largest eigenvalues.



PCA keeps the directions along which the data vary the most.

Why use singular value decomposition (SVD)?

SVD decomposes a data matrix into

$$X = UDV^T.$$

- ▶ V gives important directions in **feature space**.
- ▶ D contains singular values $s_1 \geq s_2 \geq \dots$, measuring their importance.
- ▶ U describes how the samples vary along the corresponding components.

The columns of U and V are **orthonormal**:

unit length and mutually perpendicular.

SVD reveals the dominant structure of a data matrix without first forming a covariance matrix.

How are SVD and PCA connected?

Suppose the data matrix X has been centered.
Its covariance matrix is proportional to

$$X^T X.$$

If

$$X = UDV^T,$$

then

$$X^T X = VD^2V^T.$$

Therefore:

- ▶ columns of V are the PCA directions;
- ▶ larger singular values correspond to larger variance.

PCA and SVD describe the same important directions from two closely related viewpoints.

Practical habits from Week 1

- ▶ **Check dimensions.** Make sure matrix and vector operations are mathematically valid.
- ▶ **Keep test data separate.** Do not use the test set to choose models or preprocessing.
- ▶ **Learn preprocessing from training data only.** Apply the learned transformation unchanged to validation and test data.
- ▶ **Choose evaluation measures that match the problem.** Accuracy alone may be misleading when classes are imbalanced.
- ▶ **Inspect errors, not only a final score.** Different mistakes may have different consequences.

A correct ML implementation must respect both the mathematics and the experimental design.

What a strong course project includes

A strong project contains more than a model call:

- ▶ a precise problem and defensible dataset;
- ▶ data preparation learned only from the training set;
- ▶ a meaningful comparison method and performance measure;
- ▶ controlled comparison and error analysis;
- ▶ interpretation, limitations, and possible improvements.

Week 8 will revisit these elements when we discuss project directions.

Summary

1. Machine learning connects data, models, learning, and evaluation.
2. Probability represents uncertainty; Bayes' rule reverses conditioning.
3. Expectation, variance, and covariance summarize distributions.
4. A multivariate Gaussian connects probability with data geometry.
5. Projections, eigenvectors, and SVD reveal important directions in data.

Next week: turn a probability model and observed data into an estimated parameter.

Backup

CS5487 ■ Semester A, 2026/27

Backup: course outcomes and resources

Upon successful completion of CS5487, you should be able to identify and derive common methods, implement selected methods, compare them with mathematical and experimental evidence, and design a suitable solution for a specific problem.

Course resources

Canvas contains announcements, materials, and formal deadlines. Main references are Bishop's *Pattern Recognition and Machine Learning*, Murphy's *Machine Learning: A Probabilistic Perspective*, and Duda, Hart, and Stork's *Pattern Classification*.

Backup: recommended project progression

1. **Weeks 1–2:** identify a problem, possible dataset, and initial project direction.
2. **Weeks 3–4:** define the task, evaluation measures, baseline, and experimental plan.
3. **By Week 5:** finalize the project scope and establish a working data pipeline.
4. **Weeks 6–8:** implement the baseline and begin the main experiments; refine the plan if needed.
5. **Weeks 9–12:** complete experiments, comparisons, and error analysis.
6. **Weeks 13–14:** consolidate results, discuss limitations, and prepare the presentation.

Common mistake

Start early. A strong project requires time for implementation, failed experiments, comparison, and analysis.

Backup: law of total probability

If events $\{B_k\}_{k=1}^K$ partition the sample space, then

$$\mathbb{P}(A) = \sum_{k=1}^K \mathbb{P}(A | B_k) \mathbb{P}(B_k).$$

Bayes' rule for class B_j becomes

$$\mathbb{P}(B_j | A) = \frac{\mathbb{P}(A | B_j) \mathbb{P}(B_j)}{\sum_k \mathbb{P}(A | B_k) \mathbb{P}(B_k)}.$$

Backup: a 2×2 eigendecomposition

For

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

the characteristic equation is

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (2 - \lambda)^2 - 1 = 0,$$

so $\lambda_1 = 3$, $\lambda_2 = 1$ with directions proportional to $[1, 1]^\top$ and $[1, -1]^\top$.

Probability and geometry are the foundation

Questions and discussion